

NEW PRODUCT DEVELOPMENT

A firm can obtain new products in two ways. one is through acquisition that is by ^{buying} a whole company, a patent, or a license to produce some else's product. the other is through the firm's own-product development efforts.

By new products we mean original products, product improvements, product modifications, and new brands that the firm develops through its own research and development efforts.

New products are important to both customers and the marketers who serve them. for companies, new products are a key source of growth. for customers, they bring new solutions and variety to their lives. yet innovation can be very expensive and very risky.

New products face tough odds.

According to one estimate, 90 percent of all new products fail. Each year, companies lose a huge amount of money on new product alone.

Why do so many new products fail? There are several reasons. Although an idea may be good, the

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The company may overestimate market size. The actual product may be poorly designed, or it might be - in correctly positioned, launched at the wrong time, priced too high, or poorly advertised. A high-level executive might push a favorite idea despite poor marketing research findings. Some times the costs of product development are higher than expected, and sometimes competitors fight back harder than expected. However, the reasons behind some new product failures seem pretty obvious.

Companies face a problem - they must develop new products, but the odds weight heavily against success. In add, to create successful new products, a company must understand its consumer, markets, and competitors and develop products that deliver superior value to customers. It must carry out strong new product planning and set up a systematic, customer-driven new product development process for finding and growing.

new products.

The following are the eight major steps in the new-product development process.

① Idea Generation :- New-product development starts with idea generation - the systematic search for new product ideas. A company typically generates hundreds of ideas, even thousands, in order to find a few good ones. Major sources of new product ideas include internal sources and external sources such as customers, competitors, distributors and suppliers, and others.

② Idea Screening :- The purpose of idea generation is to create a large number of ideas. The purpose of the succeeding stages is to reduce that number. The first idea-reducing stage is idea screening, which helps spot good ideas and drop poor ones as soon as possible. product development costs rise greatly in later stages, so the company wants to go ahead only with the product ideas that will

turn in to profitable products.

③ Concept Development and Testing:- An attractive idea must be developed in to a product concept. It is important to distinguish between a product idea, a product concept, and a product image. A product idea is an idea for a possible product that the company can see it self offering to the market. A product concept is a detailed version of the idea stated in meaningful consumer terms. A product image is the way consumers perceive an actual or potential product.

④ Marketing Strategy Development:- This is the process of designing an initial marketing strategy for introducing a product to the market. The marketing Strategy Statement consists of three parts. The first part describes the target market, the planned value proposition, and the sales, market share, and profit goals for the first few years. Thus.

The target market is younger, well-educated, moderate-to high income

individuals, couples, or small families seeking practical, environmentally responsible transportation.

The second part of the marketing strategy statement outlines the product's planned price, distribution, and marketing budget for the first year.

The third part of the marketing strategy statement describes the planned long-run sales, profit goals, and marketing mix strategy.

⑤ Business Analysis:- Once management has decided on its product concept and marketing strategy, it can evaluate the business attractiveness of the proposal. Business analysis involves a review of the sales, costs, and profit projections for a new product to find out whether they satisfy the company's objectives. If they do, the product can move to the product development stage.

To estimate sales, the company might look at the sales history of similar products and conduct market

Surveys. It can then estimate minimum and maximum sales to assess the range of risks. After preparing the sales forecast, management can estimate the expected costs and profits for the product, including marking R&D, operating, accounting, and finance costs. The company then uses the sales and costs figures to analyze the new product's financial attractiveness.

⑥ Product Development:- for many new-product concepts, the product may have existed only as a word description, a drawing, or perhaps a crude mock-up. If the product concept passes the business test, moves in to product development, then, Research and development (R&D) or engineering develops the product - concept in to a physical product. The product development step, however, now calls for a large jump in investment. It will show whether the product idea can be turned in to a marketable product.

The Research and development (R&D) will

department will develop and test one or more physical versions of the product concept. R&D hopes to design a prototype that will satisfy and convince consumers and that can be produced quickly and at budgeted costs. Developing a successful prototype can take days, weeks, months, or even years depending on the product and prototype methods.

⑦ Test marketing: If the product passes concept and product tests, the next is test marketing, this is the stage at which the product and marketing program are introduced in to realistic market settings.

Test marketing gives the marketer experience with marketing the product before going to the great expense of full introduction. It lets the company test the product and its entire marketing program — targeting and positioning strategy, advertising, distribution, pricing, branding and packaging and budget levels.

The amount of test marketing needed varies with each new product. Test marketing costs can be high, and

It takes time that may allow competitors to gain advantages. When the costs of developing and introducing the product are low, or when management is already confident about the new product, the company may do little or no test marketing.

⑧ Commercialization:- Test marketing gives management the information needed to make a final decision about whether to launch the new product. If the company goes ahead with commercialization.

Introducing the new product into the market - it will face high costs. The company need to build or rent a manufacturing facility. And in the case of a major new consumer packaged good, it may spend hundreds of millions of dollars for advertising, sales promotion, and other marketing efforts in the first year.

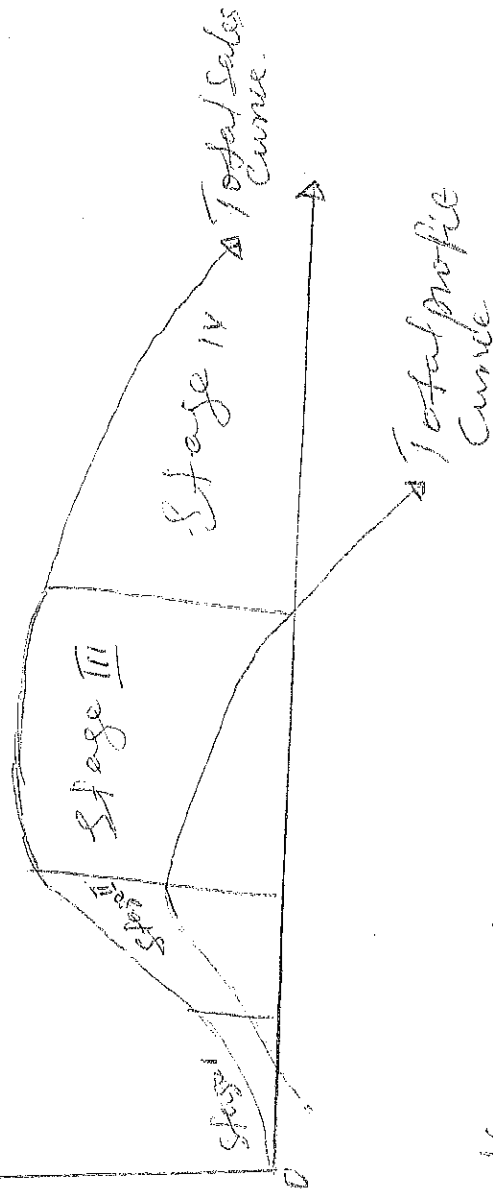
PRODUCT LIFE CYCLE

The progress made by a product in terms of its sales from the time of its market launch to that of removal from the market is referred to as 'the product life cycle'. This proposition suggests that products follow patterns of birth, the introduction stage, growth stage, maturity stage and decline stage. Like those of living organisms.

It means that all products pass through certain stages in their life time. Explaining the product life cycle concept in terms of sales, means that, a successful product from the time of its launch up to the time of its removal from the market is portrayed to have passed through

four (4) stages, as depicted in the figure below. Each stage is distinguished based on its peculiarities and characteristics.

Sales
Profit



Key: Stage I = Introduction
Stage II = Growth
Stage III = Maturity
Stage IV = Decline

1- Introduction Stage:- This is the stage when the new product is presented to the market. At this stage the market is clouded with high degree of uncertainties and risks because at the introductory stage the producer is expected to intensify his advertisement drives in order to create needs regarding the products.

This stage is characterised by low sales and revenue due to the

Increased cost. During introductory stage, it is expected that the producer should adopt one of the following strategies, rapid Skimming, Slow Skimming, rapid penetration and slow penetration. The choice of this strategy is dependent on the nature of the product, size of the market and the output presented in the market.

2- Growth Stage:- This stage is characterised by increased sales. That is, the producer will experience a colossal increase in profit via sales as a result of market acceptance. This commendable success will therefore, attract more competitors and the emergence of product Segmentation, differentiation and aggressive marketing. At this stage the producers adopt different strategies such as, enhancing product quality, increasing and redesign distribution channels and decreasing prices of products in order to attract new sets of buyers.

3- Maturity Stage:- This is the stage where the firms would start to experience decline in sales, despite the monumental increase in the total marketed sales. The decline in profit is caused by intense price competition and the attendant cost of obtaining a new buyer, which is extremely higher than the revenue expected to be realised. The stage is further identified with more distribution channels, competitors challenges, aggressive advertising, product differentiation, segmentation, distribution, and packaging. Therefore, business organisations need to adopt, product modification, market modification and marketing mix modification strategy in order to regain its status quo in the market place.

4- Decline Stage:- This is the stage where organisation experience an upsurge turn or decline in sales and consequential decrease in profit. This conspicuous decrease is precipitated by environmental changes such as, changes in

technology, changes in tastes and fashions, government policies, etc. During this stage a good number of firms withdraw from the existing market area to another due to the persistent decrease in profits. There are three common strategies that are used during the decline stage and they include maintaining strategy, harvesting strategy (reducing promotional cost to maximize profit), and deleting strategy (dropping or withdrawing the product from the market to terminate its life cycle or sell it to another firm). The measurable decrease in profit tends to force out many firms, create room for possible mergers, buy outs and ebb prices.

FORMS OF BUSINESS OWNERSHIP

There are four (4) types of business organisations that are widely in practice today and it is always advisable to evaluate the environment as well as the financial position of the proprietor before establishing any of the following firms of business organisations:-

① SOLE PROPRIETORSHIP BUSINESS:—

This is the most common form of business enterprise that is widely practiced in Nigeria. It is independently owned, operated and controlled by one person that is why it is regarded as "one man show" or "one man business". This implies that, the proprietor finances and manages the business with ^{out} seeking the assistance of another party. It is also pertinent to note that being the sole owner of the business does not necessarily mean that he

Can not employ the service of another person in the course of running the business in as much he performs the managerial responsibilities.

ADVANTAGES OF SOLE PROPRIETORSHIP BUSINESS

- 1- He is the boss.
- 2- Quick decision making.
- 3- Close relationship with customers (workers (human relation)).
- 4- Self-interest.
- 5- Small market.
- 6- Privacy.
- 7- Limited source of raw materials.

DISADVANTAGES OF SOLE PROPRIETORSHIP BUSINESS

- 1- Small capital.
- 2- Lack of continuity.
- 3- Lack of specialisation.
- 4- Unlimited liability.
- 5- Lack of skilled personnel.

MAN-MACHINE SYSTEM

Man Machine System: This is an interrelationship between man and machine before, during and after production processes in an organization. It is obvious that machine no matter how highly sophisticated cannot operate without a man to guide it. Man must set the machine before production take place, used the machine to produce something and finally keep it clean for next use. Even in automated system man is needed for surveillance. The location and design of controls, working height, information displays and the flow of work, safety features and relative utilization of both man and the machine are important determinants of quality productivity and worker acceptance of jobs. Additionally, man in some situations have to evaluate the environmental factors of heat, humidity, light, noise and hazards effects on the machine, productivity, quality and health

Production Planning and Production Control

Production planning: Production Planning is concerned with the determination, acquisition and arrangement of all facilities necessary for future operations. Similarly, is a pre-determination of manufacturing requirements such as manpower, materials, machine and manufacturing processes. It prepares a detailed plan for achieving the production goals economically, efficiently and in time. It forecasts each step in the production process. It forecasts the problems, which may arise in the production process. It tries to remove these problems. It also tries to remove the causes of wastage. Production planning and control is a tool available to management to achieve the stated objectives. Thus, a production system is encompassed by four factors i.e. quantity, quality, cost and time.

Production Control: Is the art and science of ensuring that all which occurs is in accordance with the rules established and the instructions issued. Thus, production control regulates the orderly flow of materials in the manufacturing process from

the raw material stage to the finished product. It aims at achieving production targets, optimum use of available resources, increased profits through productivity, better economic goods and services etc. An effective production control system requires reliable information, sound organization structure, a high degree of standardization and trained personnel for its successful operation.

The needs for Production planning and Control

Production planning and control are needed in an organization for the following reasons:

- ❖ Provision of raw material, equipment, machines and labor
- ❖ Reduces the production costs
- ❖ Minimize wastage of raw materials
- ❖ To organize production schedule in conformity with the demand forecasts
- ❖ Proper co-ordination of the operations of various sections/departments responsible for production.
- ❖ To ensure regular and timely supply of raw material at the desired place and of prescribed quality and quantity to avoid delays in production.
- ❖ To perform inspection of semi-finished and finished goods and use quality control techniques to ascertain that the produced items are of required specifications.
- ❖ It is also responsible for product design and development

Problems of production planning and Control in Nigeria

The problems of production planning and control centered on two major areas to include: Micro and Macro. The Problem at macro level has to do with decision of design in the following areas;

- a) Process design
- b) Work station design and equipment selection
- c) Flow of work i.e. facility planning and layout

d) Replacement policies

Problems of control at micro level have to do with design and control of systems.

- a) Materials and tools control
- b) Production control
- c) Quality controls and cost control
- d) Labor control

Quality Control

Quality control (QC) is a procedure or set of procedures intended to ensure that a manufactured product or performed service adheres to a defined set of quality criteria or meets the requirements of the client or customer. Similarly, is the process of inspecting products to ensure that they meet the required quality standards. To a manufacturer, quality is concerned with making products or providing services that are free of errors and that conform precisely to their design specification. To a user, quality may mean the product or service fits for its purposes. In sum, quality can be defined as the degree of fit between customer's expectations and customer perception of the product or services. Quality is consistent with conformance to customer's expectations.

Additionally, Quality control requires the business to create an environment in which both management and employees strive for perfection. This is done by training personnel, creating benchmarks for product quality, and testing products to check for statistically significant variations. Finally, the QC process must be ongoing to ensure that remedial efforts, if required, have produced satisfactory results and to immediately detect recurrences or new instances of trouble.

Methods of Quality Control

Below are some of the methods of Quality control in an organization: Inspection, sampling and control chart.

Inspection: This is an appraisal activity that compares goods and services to standard. Inspection can occur at three points: before production i.e., at the receiving point of raw materials, during production i.e., work-in-process or conversion and after production i.e., output inspection. Inspection deals with this basic issues; how much to inspect and how often, at what points in the process should inspection occur, whether to inspect in a centralized or onsite location and whether to inspect attributes.

Sampling plan is a blueprint for how a sampling event or program will be executed. It also a detailed outline of which measurements will be taken at what times, on which material, in what manner, and by whom. It should provide all the detail needed to ensure that representative samples are collected, handled, analyzed, and reported in a manner that meets the needs and objectives of the sampler. Implementing a clearly defined and consistently employed sampling protocol reduces the chance of error that the sampling process.

OC Curve is a graph used in quality control to determine the probability of accepting production lots when using different sampling schemes. It shows percentage-defectives along the horizontal (X), axis and probability of acceptance along the vertical (Y) axis. Lots having more than the acceptable percentage of defectives are rejected. Operating Characteristic (OC) curves are powerful tools in the field of quality control, as they display the discriminatory power of a sampling plan.

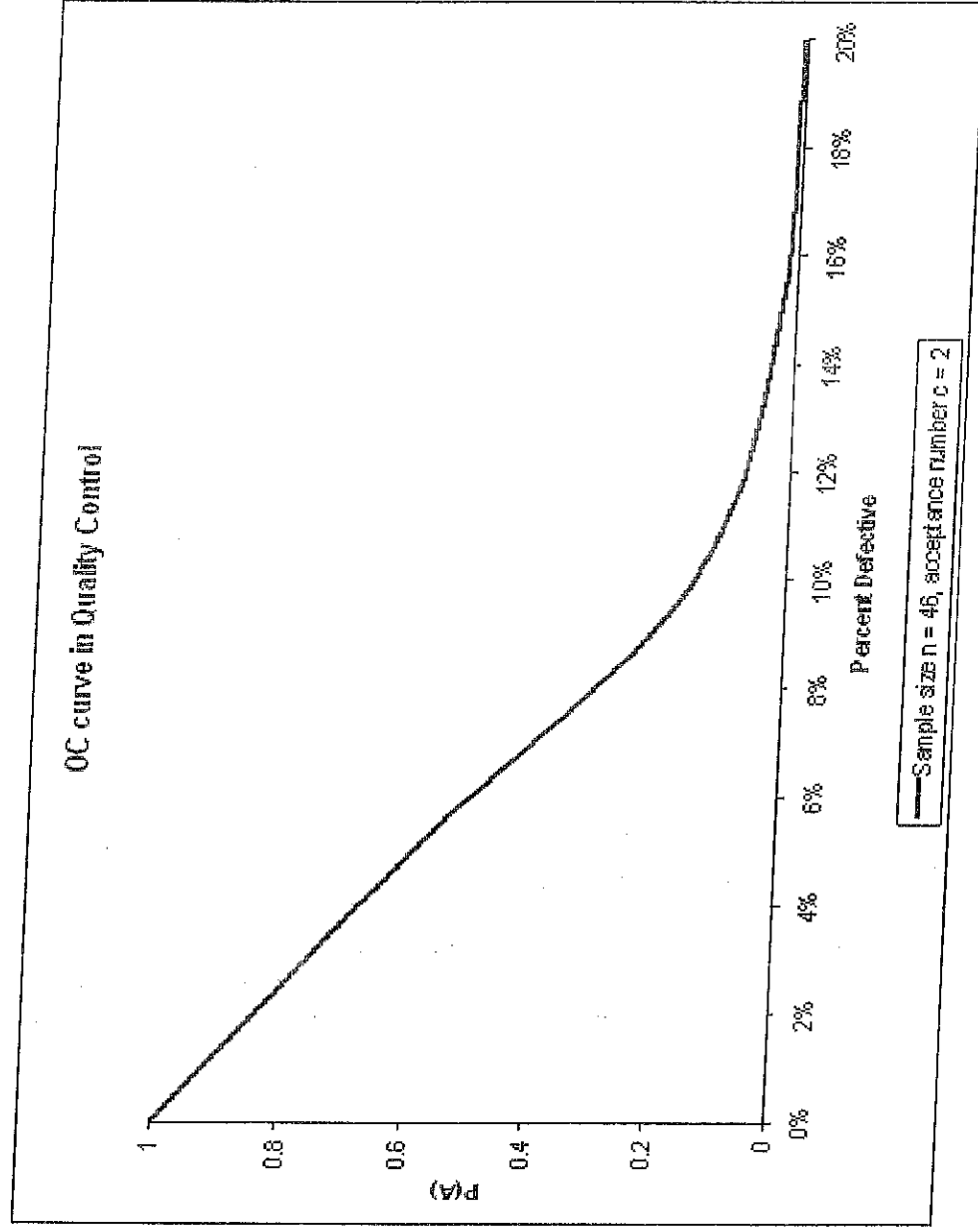


Figure 1: OC Curves in Quality Control

F- Chart is a clear graphic representation of a process from beginning to end, regardless of whether that process is an algorithm or a manufacturing procedure. The tool once used exclusively by computer programmers has now become a fundamental quality tool. Flowcharting is one of the seven management tools for quality management. Flowcharting is typically used to map a process flow showing the beginning of a process, decision points and the end of the process. If a company had to choose only one quality software, it should be a flow chart of some kind.

Example: Refers to the class illustration of F- Chart

Z Chart is a graphic chart consists of three lines which together look like the letter Z. It is prepared to represent weekly, monthly or quarterly returns. It further shows three aspects of a particular activity i.e. total for a giving period, cumulative total and moving annual total.

Maintenance

Maintenance: Maintenance can be defined as a set of activities that relate to keeping facilities and equipments in good working order and making necessary repairs when break downs occur, so that the system can perform as intended. It can also be refers to a system of maintaining and improving the integrity of production and quality system through the machines, equipment, processes, and employees that add business value to an organization.

Types of maintenance

Maintenance is of various types to include breakdown maintenance, preventive maintenance, periodic maintenance, predictive maintenance, corrective maintenance, maintenance prevention. For the purpose of this Class we consider only the preventive and corrective maintenance.

Preventive maintenance

This is a proactive approach for reducing breakdown through a program of lubrication, adjustment, cleaning, inspection and replacement of worn parts. It is a daily maintenance (cleaning, inspection, oiling and re-tightening), design to retain the healthy condition of equipment and prevent failure through the prevention of deterioration, periodic inspection or equipment condition diagnosis, to measure deterioration. The goal of preventive maintenance is to reduce incident of

breakdowns or failures in the plant or equipment to avoid the associated costs such as idle workers, schedule disruptions, injuries, damage to other equipment, products or facilities, and repairs which may involve maintaining inventories of spare parts etc. Just like human life is extended by preventive medicine, the equipment service life can be prolonged by doing preventive maintenance. It can be scheduled according to the availability of maintenance personnel and to avoid interference with operating schedules. Determining the amount of preventive maintenance to use is a function of the expected frequency of breakdown, the cost of a breakdown including actual repair costs as well as potential damage or injuries, lost production, and so on.

Example of preventive maintenance

The frequency of breakdown of machine per month is shown below. The cost of breakdown is ₦5,000 and cost of preventive maintenance is ₦ 5,500 per month. If preventive performance is performed, the probability of machine breakdown is negligible. Should the manager use preventive maintenance, or would it be cheaper to repair the machine when it breaks down? The expected number of breakdown without preventive maintenance is 1.40

No of breakdown	frequency of occurrence	expectation of occurrence
0	0.20	0
1	.30	.30
2	.40	.80
3	<u>.10</u>	<u>.30</u>
	1.00	1.40

Expected cost using repair policy is 1.40 breakdown per month x ₦5,000 breakdown = ₦ 7,000.

Preventive maintenance would cost ₦ 5,500

Therefore, preventive maintenance would yield a savings of ₦1,500/month.

Ideally, preventive maintenance would be performed just prior to breakdowns or failure because this will result in the largest possible use of facilities or equipment without a breakdown.

Corrective maintenance

Corrective maintenance is a maintenance task performed to identify, isolate, and rectify a fault so that the failed equipment, machine, or system can be restored to an operational condition within the tolerances or limits established for in-service operations. This is a maintenance which is carried out after failure detection and is aimed at restoring an asset to a condition in which it can perform its intended function. It improves equipment and its components so that preventive maintenance can be carried out reliably. Equipment with design weakness must be redesigned to improve reliability or improving maintainability. Corrective maintenance can be subdivided into

- ✓ Immediate corrective maintenance
- ✓ Deferred corrective maintenance

Maintenance plan is a systematic arrangement of important activities that relate to keeping facilities and equipments in good working order.

Maintenance Culture in Nigeria

The maintenance culture in Nigeria is very poor. That is the reason why many factories collapsed as a result of lack of preventive maintenance culture which leads to collapse of equipments, machines and the factory as a whole. To this extent, facilities that were meant to benefit the masses are just rotting away, being wasted without being put to any good use as people are not interested in making such facilities sustainable. The result therefore, is that public facilities do not last long due to lack of maintenance.

On personal levels, we must imbibe maintenance culture in order to enjoy the best value for our money, for us to enjoy the best of life, for how well we use and enjoy our buildings, machines, cars, household equipment, furniture and all other personal and public physical assets and indeed our bodies is dependent on how well we have imbibed the culture of maintenance.

Maintenance break down or preventive is very necessary for the continuing survival of a factory. It is with good maintenance culture that appropriate sound and functional machinery, equipment and tools can be maintained which would facilitate continuing operation of an industry. In a situation when parts are allowed to break down without replacement or servicing there would be system collapse which could lead to disruptions of production. If equipments are not up to date and functional quality products and or services cannot be produced.

In Nigeria, greed, non-challant attitude, corruption and some other factors seriously impede maintenance culture and hence disruptions of production activities in factories. When this occurs customers are not provided with choice and sometimes non-existence of the products or services is experienced. Employees also suffer from the effects of non-maintenance culture in Nigeria.